

Database Design Proposal for Storing Twitter Data

Name: [Harshit Shukla]

Date: 14 May 2026

Introduction

The purpose of this database design is to create a structured system for storing and managing Twitter data related to the Commonwealth Bank Twitter account.

The database is designed to store:

Original tweets posted by CommBank

User replies to tweets

Quote retweets from users

Direct mentions of the CommBank Twitter account

The proposed structure separates the information into multiple related tables to improve organization, reduce duplication, and support efficient data analysis.

Proposed Database Tables

The following tables are included in the proposed database structure:

Users

Tweets

Replies

Quote_Retweets

Mentions

Table 1: Users

This table stores information about Twitter users interacting with the CommBank account.

Variable Name	Description
user_id	Unique ID assigned to each Twitter user
username	Twitter username of the user
display_name	Public display name of the user
followers_count	Number of followers the user has
following_count	Number of accounts the user follows
account_created_date	Date the Twitter account was created
verified_status	Indicates whether the account is verified

Primary Key

user_id

Table 2: Tweets

This table stores original tweets posted by the CommBank Twitter account.

Variable Name	Description
tweet_id	Unique identifier for each tweet
user_id	ID of the user who posted the tweet
tweet_text	Content of the tweet
tweet_date	Date and time the tweet was posted
likes_count	Number of likes received
retweet_count	Number of retweets received
reply_count	Number of replies received
hashtag	Hashtags used in the tweet

Primary Key

tweet_id

Foreign Key

user_id references the Users table

Table 3: Replies

This table stores replies made by users to CommBank tweets.

Variable Name	Description
reply_id	Unique ID for each reply
tweet_id	ID of the original tweet
user_id	ID of the user posting the reply
reply_text	Content of the reply
reply_date	Date and time the reply was posted
likes_count	Number of likes on the reply

Primary Key

reply_id

Foreign Keys

tweet_id references the Tweets table

user_id references the Users table

Table 4: Quote_Retweets

This table stores quote retweets made by users.

Variable Name	Description
quote_retweet_id	Unique ID for each quote retweet

tweet_id	ID of the original tweet being quoted
user_id	ID of the user creating the quote retweet
quote_text	Text added by the user
quote_date	Date and time of the quote retweet
likes_count	Number of likes received

Primary Key

quote_retweet_id

Foreign Keys

tweet_id references the Tweets table

user_id references the Users table

Table 5: Mentions

This table stores tweets where users directly mention the CommBank account.

Variable Name	Description
mention_id	Unique ID for each mention
user_id	ID of the user making the mention
mention_text	Content of the mention
mention_date	Date and time the mention was posted
sentiment_type	Sentiment classification of the mention

Primary Key

mention_id

Foreign Key

user_id references the Users table

Table Relationships

The database tables are connected through relationships to avoid duplicate information and maintain consistency.

The relationships are as follows:

One user can create multiple tweets

One user can create multiple replies

One user can create multiple quote retweets

One user can create multiple mentions

One tweet can have multiple replies

One tweet can have multiple quote retweets

The `user_id` field connects user-related data across all tables, while the `tweet_id` field connects replies and quote retweets to the original tweet.

Potential Business Insights from Twitter Data

The proposed database structure can also support future business analytics and social media intelligence for Commonwealth Bank.

Some valuable analysis opportunities include:

Identifying which tweets receive the highest engagement through likes, replies, and retweets

Analyzing customer sentiment from replies, mentions, and quote retweets

Detecting common customer complaints or service-related issues

Understanding which topics generate the most positive customer reactions

Examining relationships between tweet views, likes, and customer interactions

Identifying peak activity times for customer engagement

Exploring audience demographics and user interaction patterns for marketing insights

These insights can help Commonwealth Bank improve customer communication, optimize marketing strategies, monitor brand reputation, and enhance customer experience through data-driven decision-making.

Conclusion

This database structure provides an organized and scalable way to store Twitter data related to the CommBank account.

The proposed design separates user information, tweets, replies, mentions, and quote retweets into independent but connected tables. This structure supports efficient querying, reduces duplicate data, and allows InsightSpark to perform future analytics such as sentiment analysis, engagement tracking, and customer behavior analysis more effectively.